Acute Inflammation

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Learning out come

- Define acute inflammation.
- Recognize causes & signs of acute inflammation.
- Identify vascular changes that occur in acute inflammation

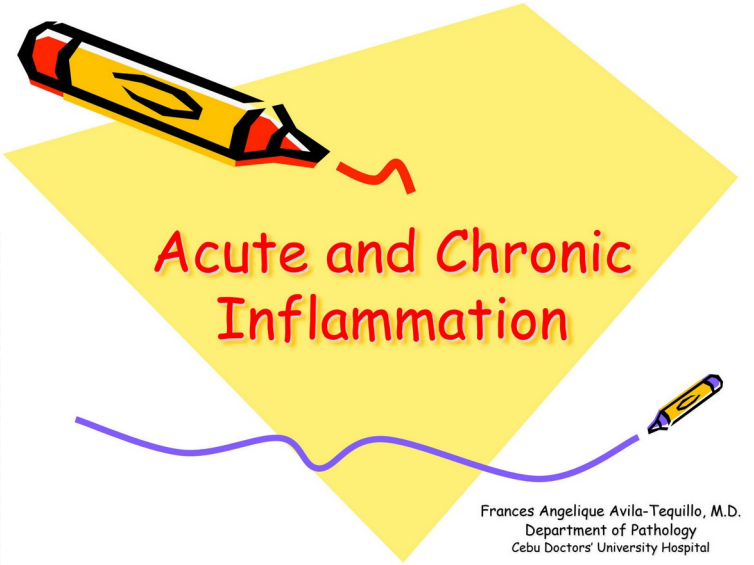
Inflammation

- Inflammation is a complex biological response of vascularized living tissue to injury.
- Aim:
- It is a protective response that is essential for survival

Purpose of inflammatory process ;

- Its primary purpose to eliminate the initial cause of cell injury, consequences of cell injury (clear out necrotic cells and tissues), and initiate the process of healing which always must follow any inflammatory process .

Inflammation



Inflammation is of 2 types:

- **Acute Inflammation**
  - ✓ due to early response by the body
  - ✓ short duration
- **Chronic Inflammation**
  - ✓ occurs after delay
  - ✓ it is for longer duration
  - ✓ Characterised by response by chronic inflammatory cells.

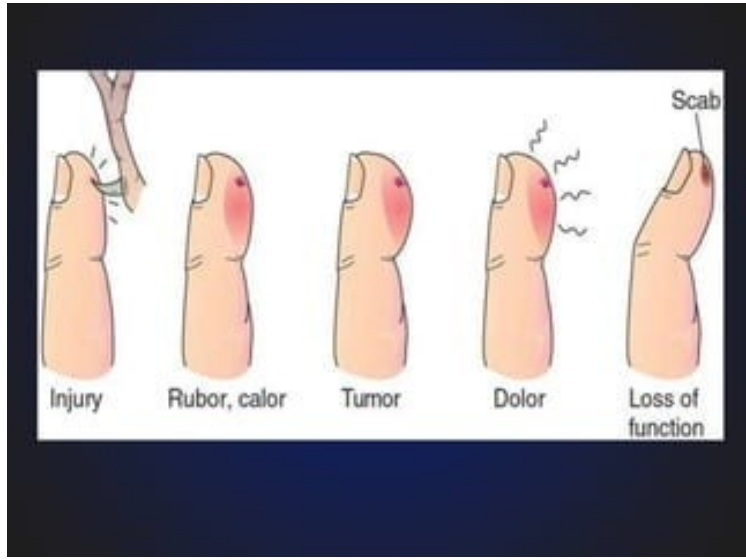
Trigger Stimuli For Acute Inflammation

Stimuli	Description
Infections	Bacteria, viruses, fungi, parasites
Trauma	Blunt and penetrating trauma: examples include burns, frostbite, irradiation and environmental chemicals
Tissue necrosis	Examples of things that trigger necrosis include ischemia from a heart attack and physical/chemical injuries
Foreign bodies	Foreign bodies include splinters, sutures and dirt
Immune (hypersensitivity reactions)	Are triggered either against environmental substances or the body attacks itself (this process is known as autoimmune diseases: auto- meaning 'self')

Classical (cardinal) clinical sings of inflammation:

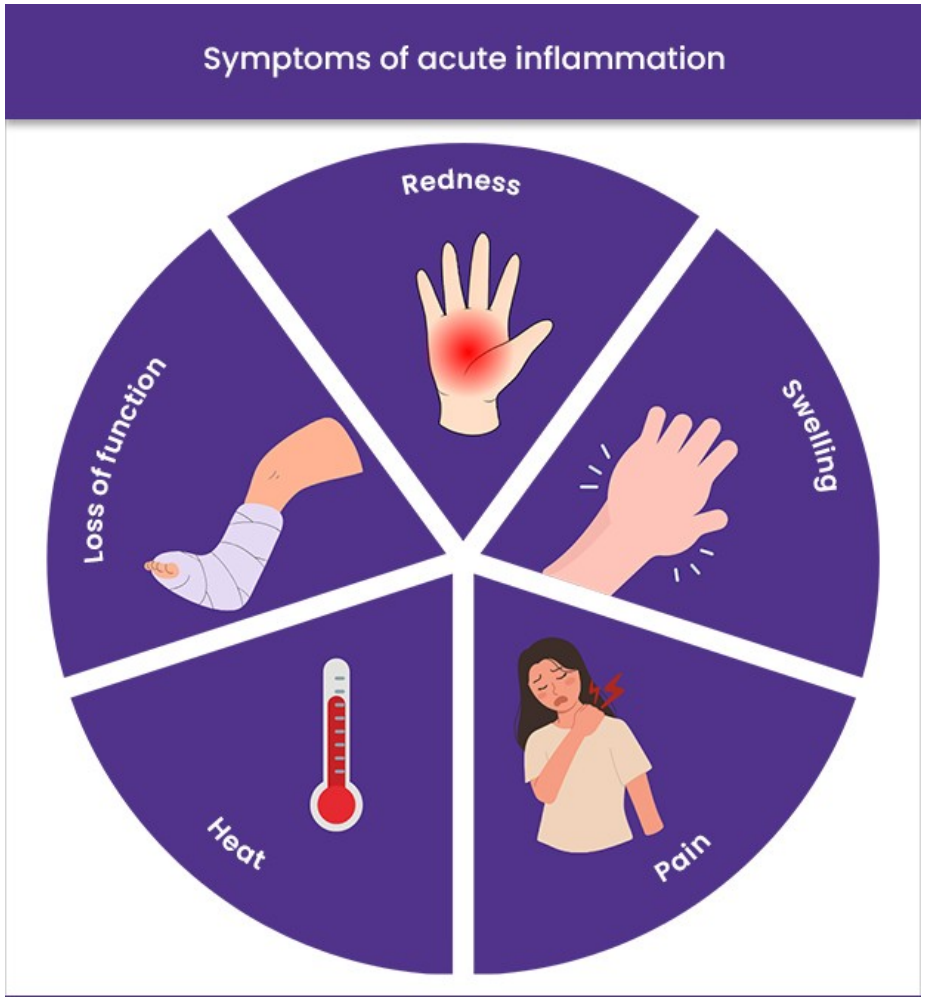
❖There are 5 cardinal signs of inflammation (most prominent in acute inflammation):

- **Heat** (Latin: calor) due to vascular dilation
- **Erythema** (Latin: rubor) due to vascular dilation and congestion
- **Edema** (Latin: tumor) due to increased vascular permeability
- **Pain** (Latin: dolor) due to mediator release
- **Loss of function** (Latin: functio laesa) due to pain, edema, tissue injury, and/or scar



Cardinal sign of acute inflammation





I
n general, adding The suffix (itis) to the name of organs signifies acute inflammation, (although there are exceptions like pneumonia*)*
e.g. colitis, appendicitis , and mastitis.

Organ/tissue	Inflammatory disease
• Tongue	Glossitis
• Oral mucosa	Stomatitis
• Cornea	Keratitis
• Lung (alveolar spaces)	Pneumonia
• Lung (interstitium)	Pneumonitis
• Stomach	Gastritis
• Small intestine	Enteritis
• Cecum	Typhlitis
• Rectum	Proctitis
• Testicle	Orchitis
• Vagina	Colpitis
• Fallopian tube	Salpingitis
• Navel	Omphalitis
• Spleen	Splenitis
• Breast	Mastitis
• Adipose tissue	Panniculitis
• Brain parenchyma	Encephalitis

ACUTE INFLAMMATION

VASCULAR EVENTS

1. HAEMODYNAMIC CHANGES
2. ALTERED VASCULAR PERMEABILITY

CELLULAR EVENTS

1. EXUDATION OF LEUKOCYTES
2. PHAGOCYTOSIS



Main steps of inflammation

- **Vascular events:**
  - **Vasodilation**
  - Increase of vascular permeability
  - Formation of **exudate**
- **Cellular events:**
  - Margination, rolling, adhesion
  - Transmigration (diapedesis)
  - Chemotaxis
  - Activation of neutrophilic granulocytes
  - Phagocytosis: recognition, binding, engulfment, destruction
- **Resolution**
- The outcome can be: healing, formation of scar tissue, chronic inflammation



Chemical mediators

Vascular events

Initial transient vasoconstriction of arterioles.

↓

Persistent progressive vasodilatation.

↓

Elevation of the local hydrostatic pressure.

↓

Increase in vascular permeability.

↓

Transudation of fluid into the extracellular space.

↓

Slowing or stasis of microcirculation.

↓

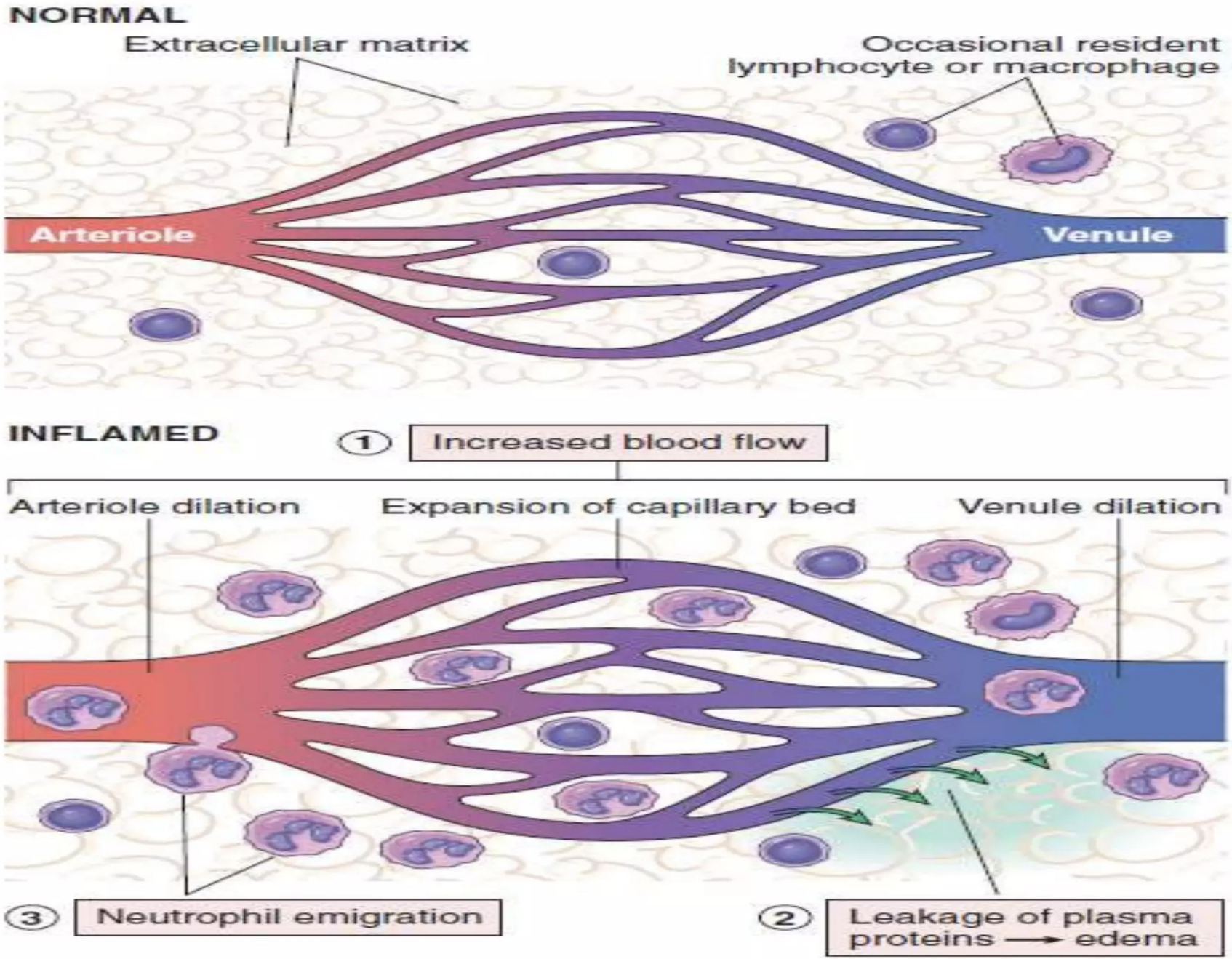
Leucocytic margination.

A/ Change in vascular flow & caliber

Initial transient vasoconstriction followed by

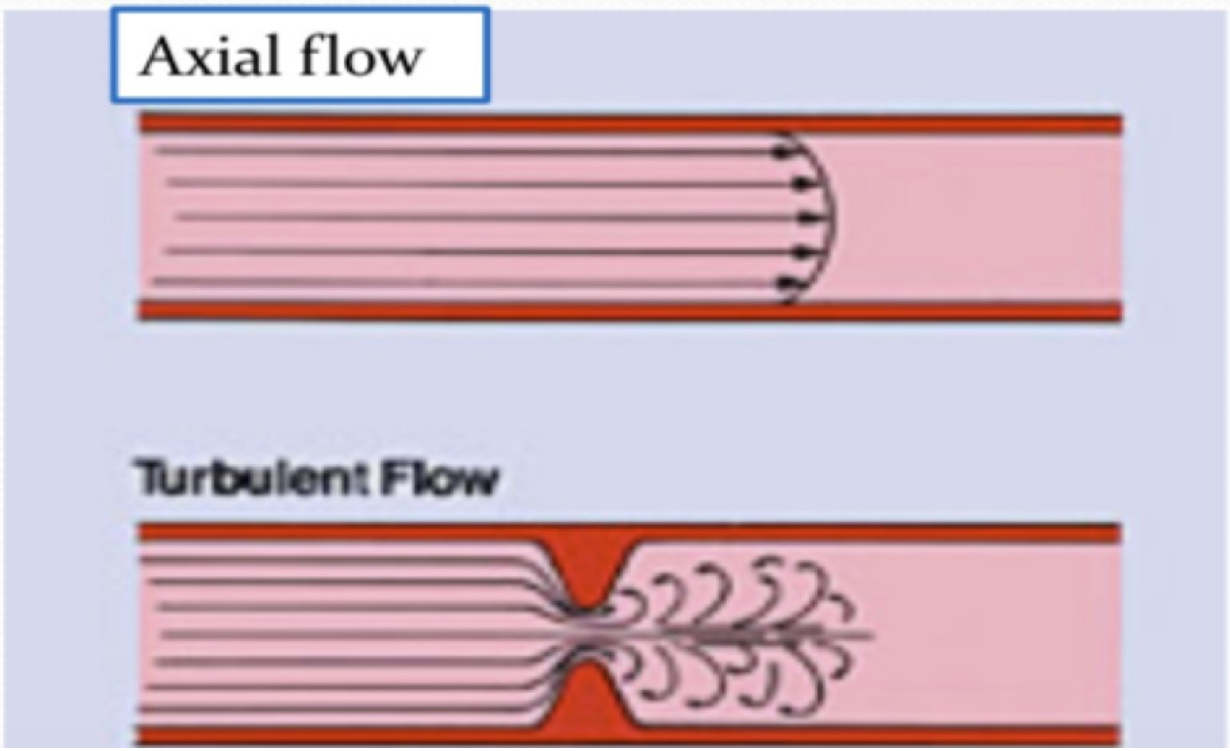
- vasodilation which is induced by the action of several mediators, mainly histamine. This result in redness and heat
- Vasodilation is quickly followed by increased permeability of the microvasculature, with outpouring of protein-rich fluid into the extravascular tissues(odema)

- stasis The loss of fluid and increased vessel diameter lead to slower blood flow, increased concentration of red cells in small vessels, and increased viscosity of the blood. These changes result in the engorgement of small vessels with slowly moving red cells, a condition termed (numerous dilated small vessels packed with RBCs
- After stasis - (the cellular event will start)-leukocytes (Neutrophils) change their normal flow in vessels(normally in the center), & begin to accumulate along the vascular endothelial surface


NORMAL
Extracellular matrix
Occasional resident lymphocyte or macrophage
Arteriole
Venule
INFLAMED
1 Increased blood flow
Arteriole dilation
Expansion of capillary bed
Venule dilation
3 Neutrophil emigration
2 Leakage of plasma proteins → edema

Changes in blood flow

- **Turbulent flow- disturbed axial flow**
  - The blood cells especially white cells come in to contact with vessel wall and making adhesions between vessel wall and WBCs.



B- Increased Vascular Permeability (Vascular Leakage): B- Increased Vascular Permeability (Vascular Leakage):

B- Increased Vascular Permeability (Vascular Leakage):

- Several mechanisms
- 1-Retraction of endothelial cells : most common mechanism
- resulting in the opening of inter-endothelial spaces . It is elicited by histamine, bradykinin, leukotrienes, and other chemical mediators.
- It is called the immediate transient response because it occurs rapidly after exposure to the mediator and is usually short-lived (15 to 30 minutes).
- This is to be distinguished from the delayed prolonged response that follows an endothelial injury.

- 2- Endothelial injury resulting in endothelial cell necrosis & detachment.
- It is encountered in severe injuries, for example, in burns, or is induced by the actions of microbes and microbial toxins that target endothelial cells.
- Neutrophils that adhere to the endothelium during inflammation may also injure the endothelial cells and thus amplify the reaction.
- In most instances leakage starts immediately after injury and is sustained for several hours until the damaged vessels are thrombosed or repaired

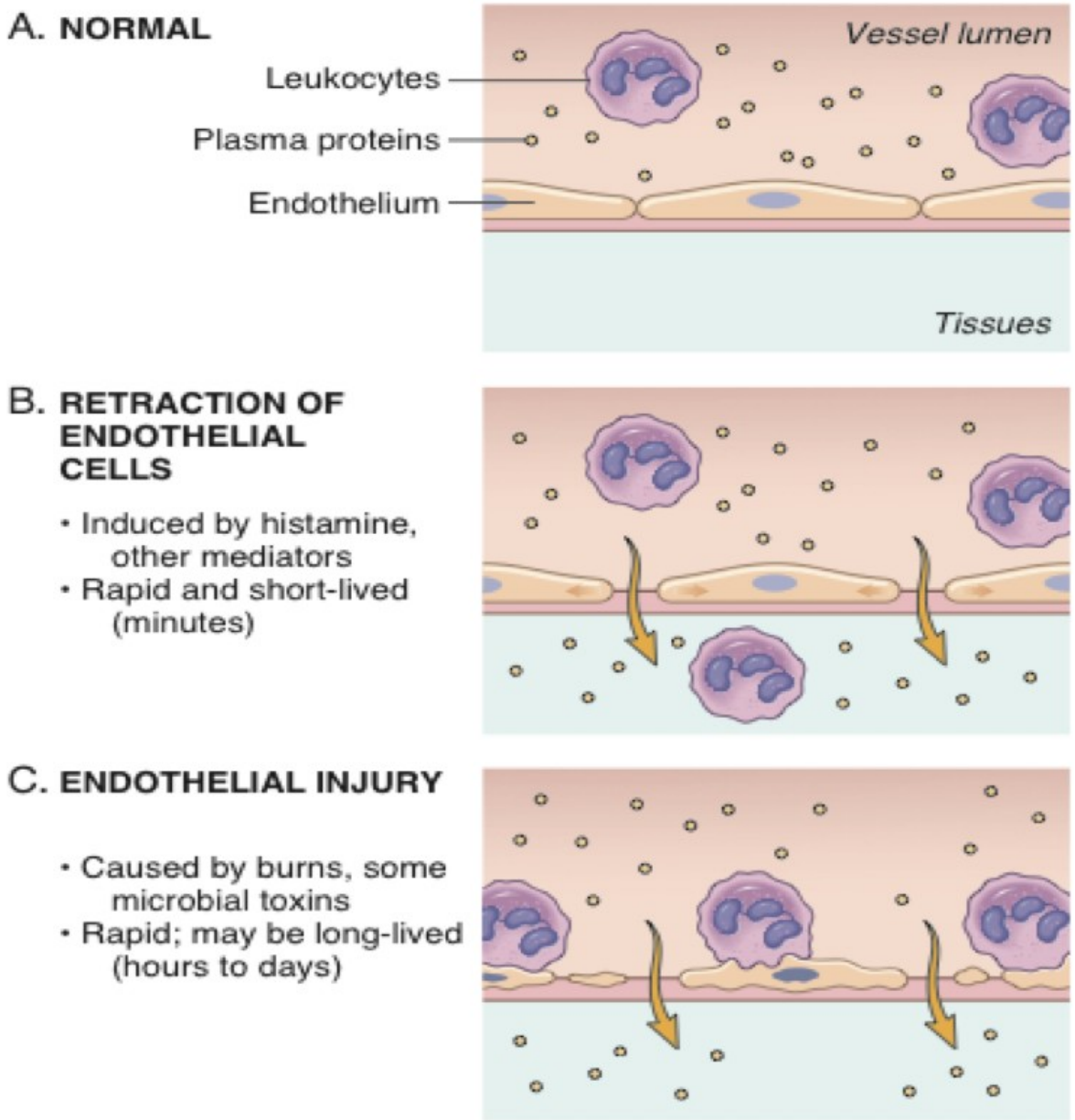
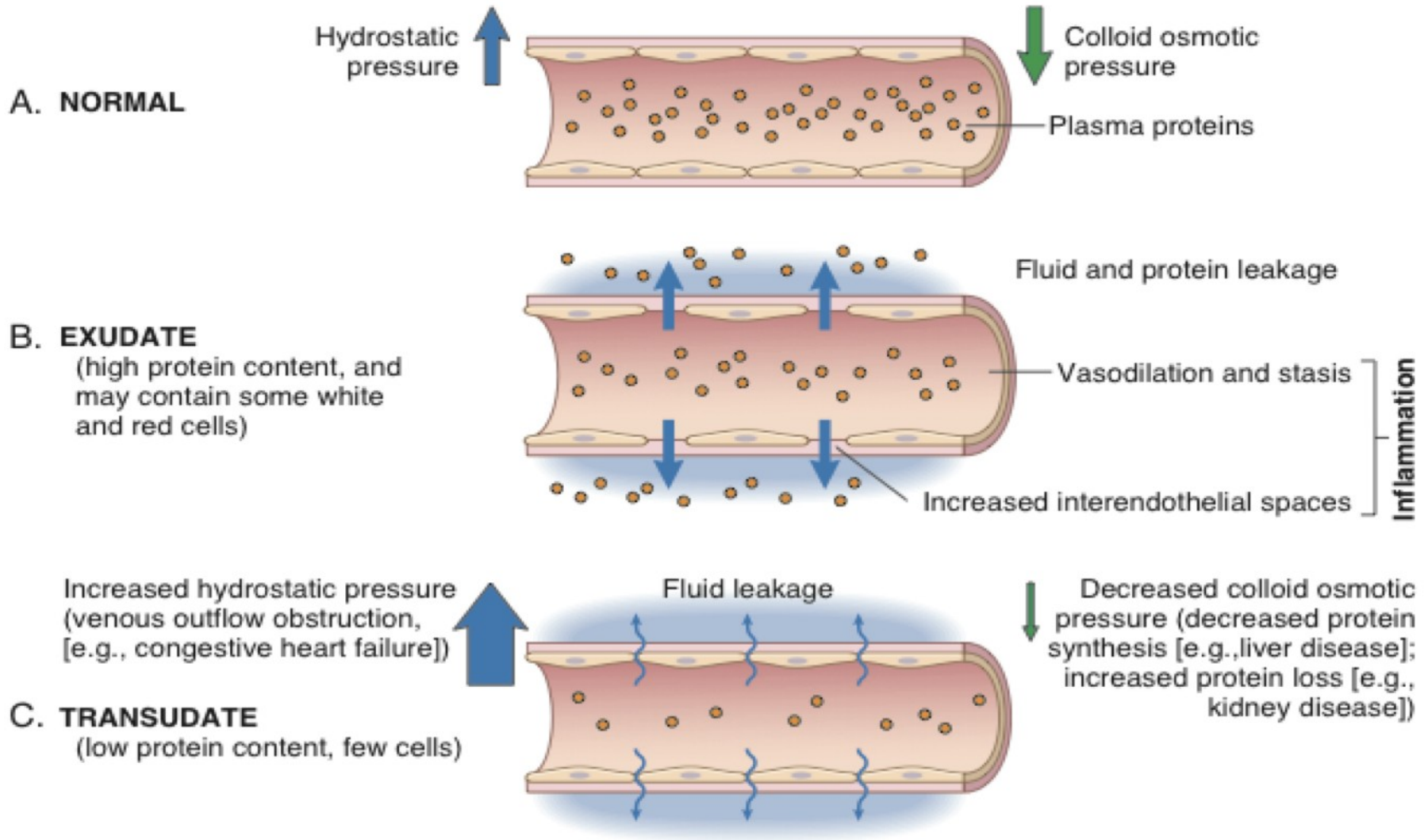


Figure 3-3 Principal mechanisms of increased vascular permeability in inflammation and their features and underlying causes.

- Odema :
- Excess fluid in the interstitial tissue or serous cavities , can be either exudate or transudate
- Exudate : In Inflammation / infection
- (v. permeability)
- LEAKAGE OF HIGH PROTEINACEOUS FLUID, with high cellularity
- Transudate : in contrast to exudate , Non inflammatory odema :
- it is essentially an ultra-filtrate of blood plasma result in
- ⇈ hydrostatic pressure
  ↓colloid pressure,
- LEAKAGE OF LOW PROTEINACEOUS FLUID, with low cellularity
-

Formation of exudates & transudates.

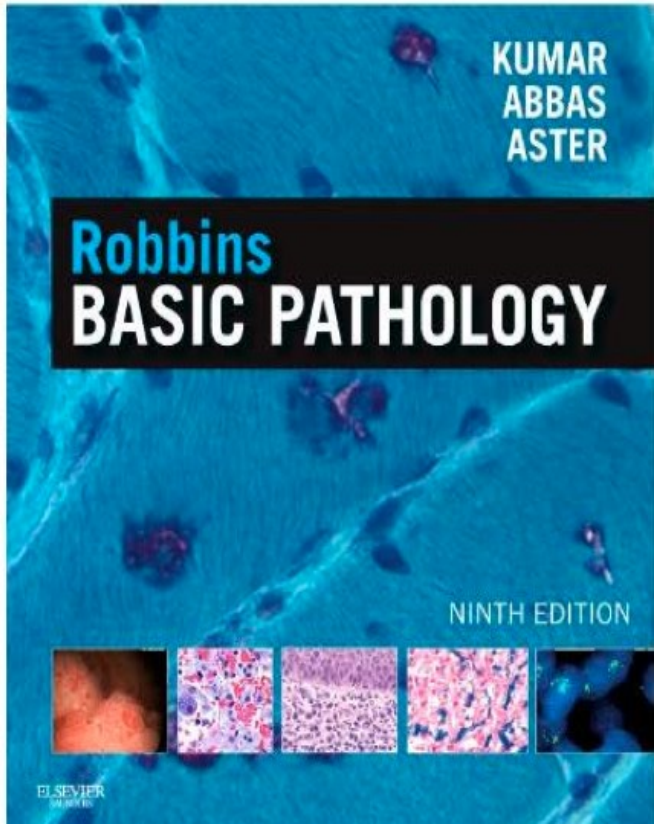


Transudate Vs Exudate

Transudate	Exudate
Filtrate of blood plasma without changes in endothelial permeability	Oedema of inflamed tissue with increased vascular permeability
Non-Inflammatory Oedema	Inflammatory Oedema
Protein < 1g/dl	Protein – High 2.5-3.5 g/dl
Glucose – same as plasma	Glucose – Low (<60 mg/dl)
Specific Gravity < 1.015	Specific Gravity > 1.018
pH > 7.3	pH < 7.3
Few Cells, mainly mesothelial cells and cellular debris	Inflammatory as well as parenchymal cells
eg. Oedema in congestive cardiac failure	Purulent exudate such as pus

Reference

- Recommended reference book
  Chapter 2



Thank You